

Software Requirements Proposal

Multi-Level Virtual Points Gaming Platform

Version: 1.0

Prepared For: Client

Prepared By: Development Team

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Executive Summary

This document presents the initial functional requirements and proposed solution for the development of a **Multi-Level Virtual Points Gaming Platform**.

The platform is designed to provide a secure, role-based environment where authorized users manage account creation, distribute virtual points, and control access to a game. The system emphasizes accountability, security, scalability, and complete transaction traceability.

The purpose of this document is to establish a shared understanding of the project's objectives, workflow, features, and development scope before implementation begins.

1. Project Objective

The objective of the project is to develop a secure web application that enables hierarchical management of users and virtual points while allowing only authorized players to participate in the game.

The platform will provide:

- Controlled account creation
 - Secure authentication
 - Hierarchical user management
 - Virtual point distribution
 - Game access management
 - Administrative reporting
 - Comprehensive audit tracking
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2. Business Overview

The platform follows a hierarchical organizational model.

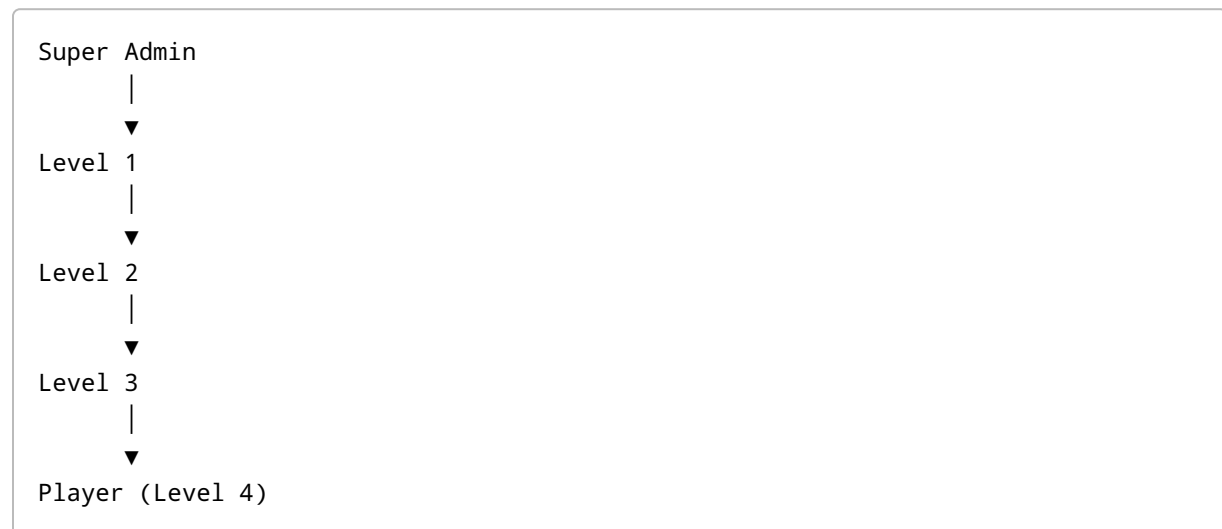
Only authorized users may create accounts beneath them.

Virtual points move through the hierarchy until they reach players, who use those points to access the game.

Every activity within the system is recorded to ensure transparency and accountability.

3. System Architecture

User Hierarchy



Each role has clearly defined responsibilities and permissions.

4. User Roles & Responsibilities

Super Administrator

The Super Administrator has unrestricted access to the platform.

Responsibilities include:

- Create Level 1 accounts

- Monitor the complete hierarchy
 - Manage platform configuration
 - Adjust virtual point balances
 - Freeze or activate user accounts
 - Access reports and analytics
 - Configure game settings
 - Review audit logs
 - Manage system security
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Level 1

Responsibilities:

- Create Level 2 users
 - Transfer virtual points
 - Monitor assigned Level 2 users
 - Review transaction history
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Level 2

Responsibilities:

- Create Level 3 users
 - Transfer virtual points
 - Manage assigned Level 3 users
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Level 3

Responsibilities:

- Create Player accounts
 - Allocate points
 - Monitor player activity
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Player (Level 4)

Responsibilities:

- Secure login
- View wallet balance
- Access the game

- View gameplay history

Players cannot create accounts or transfer points.

5. User Registration Workflow

Public registration is not available.

Accounts are created exclusively through the management hierarchy.

Registration flow:

- Super Admin creates Level 1
- Level 1 creates Level 2
- Level 2 creates Level 3
- Level 3 creates Players

This model ensures controlled growth and clear ownership of every account.

6. Virtual Point Management

Virtual points represent access to gameplay.

Point distribution follows the management hierarchy.

```
graph TD; SA[Super Admin] --> L1[Level 1]; L1 --> L2[Level 2]; L2 --> L3[Level 3]; L3 --> P[Player]; P --> G[Game];
```

Business Rules:

- Points move only downward.
- Points cannot skip hierarchy levels.
- Every transfer is permanently recorded.

- Players consume points when accessing the game.
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7. Gameplay Workflow

The gameplay process consists of the following steps:

1. Player logs into the platform.
 2. Wallet balance is verified.
 3. Required points are deducted.
 4. Game session begins.
 5. Score and gameplay information are recorded.
 6. Reports are updated automatically.
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8. Functional Modules

Authentication

- Secure login
 - Password reset
 - Session management
 - Role-based authorization
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User Management

- Account creation
 - Profile management
 - Password management
 - Freeze and activate users
 - Hierarchy management
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Wallet Management

- Virtual balance
 - Point transfers
 - Transaction history
 - Balance monitoring
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Game Management

- Game access
 - Score recording
 - Gameplay history
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Reporting

The system will generate:

- Daily reports
- Weekly reports
- Monthly reports
- Point distribution reports
- Player activity reports
- Login reports

Reports will support export to CSV and PDF.

Notification System

Users receive notifications for significant events, including:

- Account creation
 - Point transfers
 - Password changes
 - Account status updates
 - Game completion
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Audit Logging

The system maintains immutable logs for:

- User creation
 - Point transfers
 - Administrative actions
 - Login activity
 - Password resets
 - Game sessions
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9. Administrative Dashboard

The administrative dashboard will provide centralized platform management.

Key capabilities include:

- User statistics
 - Active user monitoring
 - Complete hierarchy visualization
 - Point distribution tracking
 - Transaction monitoring
 - Analytics dashboard
 - Audit logs
 - Security management
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10. Security Requirements

Security is a core component of the platform.

The solution will include:

- Role-Based Access Control (RBAC)
 - Secure password hashing
 - JWT-based authentication
 - Session timeout
 - Login history
 - Account lockout protection
 - Audit trail
 - Permission validation on all operations
-

11. Non-Functional Requirements

The platform should be:

- Secure
- Scalable
- Responsive
- Maintainable
- Reliable
- Modular
- Extensible

The architecture should support future enhancements without significant redesign.

12. Proposed Technology Stack

Frontend

- React
- Tailwind CSS

Backend

- Node.js
- Express.js

Database

- PostgreSQL

Authentication

- JWT

Version Control

- Git
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13. Project Deliverables

The completed solution will include:

- Responsive web application
 - Administrative dashboard
 - Role-based authentication
 - User management system
 - Virtual wallet
 - Point transfer system
 - Game integration
 - Reporting module
 - Notification module
 - Audit logging
 - API documentation
 - Deployment support
 - Source code
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14. Assumptions

The following assumptions apply:

- Virtual points are used only within the platform.
 - The client will provide game-specific rules and assets if required.
 - Hosting infrastructure and domain registration are outside the scope unless otherwise agreed.
 - Changes beyond the approved requirements may require a revised timeline and budget.
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15. Project Phases

Phase	Deliverable
Phase 1	Requirements validation and planning
Phase 2	UI/UX design and database architecture
Phase 3	Backend API development
Phase 4	Frontend implementation
Phase 5	Game integration
Phase 6	Testing and quality assurance
Phase 7	Deployment and user acceptance

16. Success Criteria

The project will be considered successful when:

- All user roles function according to defined permissions.
 - Virtual point transfers are secure and accurately recorded.
 - Only authorized players can access the game.
 - Administrative reporting is accurate and complete.
 - The platform operates reliably under expected usage.
 - All critical actions are fully auditable.
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Conclusion

This proposal outlines a secure, scalable, and maintainable platform designed to manage hierarchical users, virtual point distribution, and controlled game access. The proposed architecture emphasizes security,

operational transparency, and future extensibility, providing a strong foundation for long-term growth and additional functionality.